Direct Strength Method for Design of Cold-Formed Steel Columns with Holes

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ABSTRACT

In this paper design expressions are derived that extend the American Iron and Steel Institute (AISI) Direct Strength Method (DSM) to cold-formed steel columns with holes. For elastic buckling controlled failures, column capacity is accurately predicted with existing DSM design equations and the cross-section and global elastic buckling properties calculated including the influence of holes. For column failures in the inelastic regime, where strength approaches the squash load, limits are imposed to restrict column capacity to that of the net cross section at a hole. The proposed design expressions are validated with a database of existing experiments on cold-formed steel columns with holes and over 200 nonlinear finite element simulations which evaluate the strength prediction equations across a wide range of hole sizes, hole spacings, hole shapes, and column dimensions. The recommended DSM approach is demonstrated to provide a broad improvement in prediction accuracy and generality when compared to the AISI Main Specification, and with the recent introduction of simplified methods for calculating elastic buckling properties including the influence of holes, is ready for implementation in practice.

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INTRODUCTION

Cold-formed steel is a popular engineered material in residential and commercial construction because of its inherent structural efficiency gained through cold-bent curvature and its broad spectrum of prefabricated geometries. The thin-walled structural steel members are manufactured at a roll-forming plant, where steel sheet is cold-bent, typically into an open cross-section, for example a C- or Z-section. Near the end of the assembly line, holes are punched with a hydraulic die to accommodate electric and plumbing conduits, as in, the lipped C-section structural stud column in Fig. 1(a). Web holes also serve as intermediate brace connection points in structural stud walls [Fig. 1(b)], and with recent advances in machining equipment, roll-forming manufacturers can provide custom solutions with intricate hole shapes and patterns [Fig. 1(c)].

(a) (b) (c)

Fig. 1. Examples of holes in cold-formed steel: (a) punched holes to accommodate utilities, (b) a web hole is used as a wall bracing access point, (c) complex hole geometries (photo courtesy of SADEF N.V.)

The broad range of hole shapes, sizes, and spacings in cold-formed steel construction today is exceeding the original scope of the American Iron and Steel Institute (AISI) design equations developed for columns with holes over the last four decades. The current AISI design equations were derived within the context of the

effective width method (Peköz 1987), which accounts for the influence of holes on local buckling dominated failures over a narrow range of hole sizes, shapes, and spacings. For example, AISI strength prediction equations for a stiffened element (e.g., the web of a C-section) with non-circular holes is limited to a centerline spacing of 600 mm or greater, and the width of a hole must be less than 63 mm regardless of the column length or cross-section dimensions, reflecting the empirical extension of the effective width method based on selected testing.

The AISI specification addresses the influence of holes on local buckling through the effective width method, however the presence of holes is not currently considered for global buckling or distortional buckling-controlled failures. When holes are present in a cold-formed steel column, the critical elastic flexural and flexural-torsional (global) buckling loads are lower than the same column without holes, which increases the global slenderness and decreases predicted strength (Moen and Schafer 2009a). Considering distortional buckling, a form of buckling related to intermediate and/or edge stiffeners commonly observed in open cross-sections, the presence of web holes decreases the stabilizing influence of the web on the cross-section, reducing the critical elastic distortional buckling load and increasing the tendency for distortional buckling to initiate at a hole (Kesti 2000; Moen and Schafer 2008; Moen and Schafer 2009a). A more general cold-formed steel design method that considers the influence of holes across all strength limit states is needed.

The AISI research program summarized herein capitalizes on recent advances in cold-formed steel strength prediction, and specifically the implementation of the AISI Direct Strength Method (DSM) (AISI-S100 2007, Appendix 1). DSM represents an

important advancement in cold-formed steel design because it provides engineers and cold-formed steel manufacturers with the tools to predict member strength with any general cross-section. This research extends the appealing generality of the DSM approach to cold-formed steel members with holes, resulting in a design method that can accommodate the expanding range of hole sizes, shapes and spacings employed in industry.

STRATEGY FOR EXTENDING DSM TO COLUMNS WITH HOLES

The AISI Direct Strength Method employs the elastic buckling properties of a general cold-formed steel cross-section to predict strength. For members without holes, the elastic buckling properties are obtained from an elastic buckling curve generated with freely available software, for example CUFSM (Schafer and Ádàny 2006), that performs a series of eigen-buckling analyses over a range of buckled half-wavelengths. An elastic buckling curve is provided in Fig. 2 for a cold-formed steel C-section column. The critical elastic buckling loads for local and distortional buckling, i.e. $P_{cr\ell}$ and P_{crd}, are defined by the local minima on the design curve and the global (Euler) buckling load, P_{cre}, is read off the curve at the effective length of the member (Li and Schafer 2010). The buckling loads are input into DSM design expressions to calculate the column strength (AISI-S100 2007, Appendix 1). The global buckling capacity, P_{ne}, is:

$$\text{for } \lambda_c \le 1.5,\ P_{ne} = \left(0.658^{\lambda_c^2}\right)P_y$$

$$\text{for } \lambda_c > 1.5,\ P_{ne} = \left(\frac{0.877}{\lambda_c^2}\right)P_y \qquad (1)$$

where $\lambda_c = (P_y / P_{cre})^{0.5}$. The local buckling capacity, $P_{n\ell}$, is:

$$\text{for } \lambda_\ell \le 0.776\,,\ P_{n\ell}=P_{ne}$$

$$\text{for } \lambda_\ell > 0.776\,,\ P_{n\ell}=\left[1-0.15\left(\frac{P_{cr\ell}}{P_{ne}}\right)^{0.4}\right]\left(\frac{P_{cr\ell}}{P_{ne}}\right)^{0.4} P_{ne} \qquad (2)$$

where $\lambda_\ell=(P_{ne}/P_{cr\ell})^{0.5}$. The distortional buckling capacity, P_{nd}, is:

$$\text{for } \lambda_d \le 0.561\,,\ P_{nd}=P_y$$

$$\text{for } \lambda_d > 0.561,\ P_{nd}=\left(1-0.25\left(\frac{P_{crd}}{P_y}\right)^{0.6}\right)\left(\frac{P_{crd}}{P_y}\right)^{0.6} P_y \qquad (3)$$

where $\lambda_d=(P_y/P_{crd})^{0.5}$. The column capacity, P_n, is taken as the minimum strength of the local, global, and distortional limit states, i.e. $P_n=\min(P_{n\ell}, P_{nd}, \text{and } P_{ne})$.

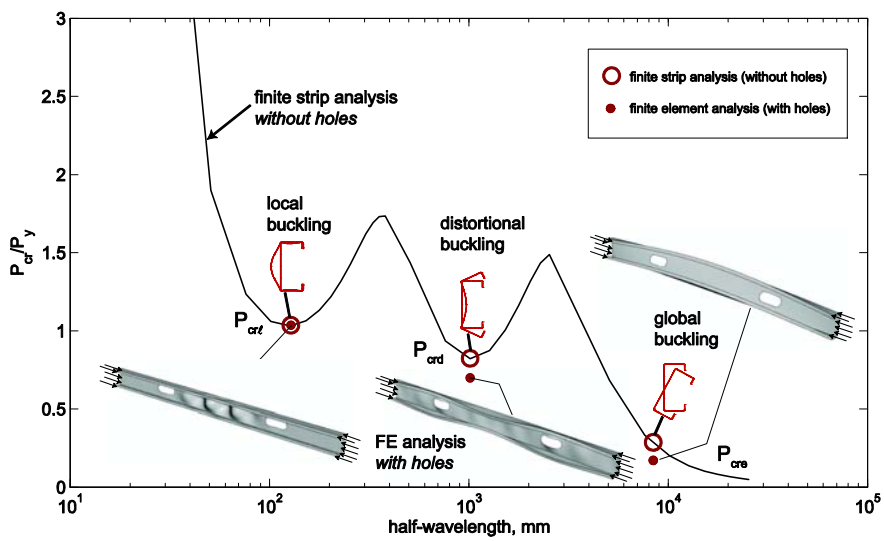

Fig. 2 The column elastic buckling curve, generated with a finite strip analysis, can be used to obtain the local, distortional, and global buckling loads. FE eigen-buckling analysis of a column with holes is provided for comparison.

A natural extension of the Direct Strength approach to columns with holes is to maintain the assumption that elastic buckling properties are viable parameters for predicting member strength. For a column with holes, this means that the elastic buckling loads $P_{cr\ell}$, P_{crd}, and P_{cre}, are calculated including the influence of holes, and the

buckling loads are input into DSM design expressions. But how will engineers calculate the elastic buckling loads including the influence of holes? The finite strip method, as employed in freely available programs such as CUFSM for members without holes, cannot accommodate discrete holes along the length of a column in an elastic buckling analysis. Thin shell finite element eigen-buckling analysis is a viable option (see Fig. 2), although meshing around holes can be time consuming and the local, distortional, and global buckling loads must be manually identified from 1000's of modes.

To address the challenge of quantifying elastic buckling for cold-formed steel columns with holes, a suite of simplified methods was recently developed as a convenient alternative to thin shell finite element eigen-buckling analysis. Engineering expressions for the elastic buckling of thin plates with holes are now available for stiffened and unstiffened elements (Moen and Schafer 2009b). Hand methods and new procedures utilizing the finite strip method can approximate the local, distortional, and global buckling loads of cold-formed steel columns with holes for holes shapes, sizes, and spacings common in industry (Moen and Schafer 2009a). An example elastic buckling calculation utilizing the simplified methods is presented for a structural stud with holes in Moen and Schafer (2010), where auxiliary finite strip analysis is performed at the net section, along with hand calculations that approximate the reduced global buckling properties.

With the elastic buckling infrastructure in place, research efforts have shifted to the load-deformation response and failure modes of cold-formed steel columns with holes, the goal being to identify what changes are required to the existing DSM design expressions to extend their viability to columns with holes. The following sections

describe these efforts, starting with the assembly of a database containing strengths and elastic buckling properties of cold-formed steel columns with holes tested over the past 30 years. The elastic buckling loads are obtained with thin shell finite element eigen-buckling studies that include the influence of holes and experimental boundary conditions on elastic buckling behavior. A portion of the database is utilized to validate a nonlinear finite element modeling protocol for cold-formed steel columns (Moen 2008), which is then implemented to explore strength trends across a wide range of global and cross-sectional slenderness, hole size and hole spacings. The finite element studies reveal how holes influence distortional buckling and local-global buckling interaction failure modes, and are used to guide the development of the proposed DSM expressions for cold-formed steel columns with holes presented at the conclusion of this manuscript.

EXPERIMENTAL DATABASE OF CFS COLUMNS WITH HOLES

Historically, experiments on cold-formed steel compression members with holes have focused on stub column tests. These stub column tests were used to develop and validate the current AISI effective width design equations for the unstiffened strips on either side of a hole (Abdel-Rahman and Sivakumaran 1998; Miller and Peköz 1994; Ortiz-Colberg 1981; Pu et al. 1999; Sivakumaran 1987). A small number of pinned-pinned long column tests with holes have also been conducted to explore global buckling failures (Ortiz-Colberg 1981). Recent experiments performed by the authors quantified the influence of slotted holes on distortional-buckling controlled failures of short and intermediate length lipped C-section columns (Moen and Schafer 2008). Results from these studies were assembled in a database to serve as a source of validation supporting the proposed DSM design expressions for columns with holes.

The experimental database includes tested strengths for 78 column specimens. Accompanying the test results in the database are the local ($P_{cr\ell}$), distortional (P_{crd}), and global buckling loads (P_{cre}), including the influence of holes and experimental boundary conditions, which were calculated for each specimen with thin shell finite element eigen-buckling analysis in ABAQUS (ABAQUS 2007). The range of cross-section and hole dimensions contained in the database are provided in Table 1, with Fig. 3 defining the column geometry notation. All column specimens are lipped C-sections meeting DSM prequalification standards (AISI-S100-07, Appendix 1). Complete details of the database development, including member dimensions, material properties, and boundary conditions are summarized in Moen (2008).

Table 1. Experimental database of cold-formed steel columns with holes

					H/t		*B/t*		*D/t*		*H/B*		*L/H*		h_{hole}/H		S/L_{hole}	
	End conditions	Hole shape	Length	Count	min	max	min	max	min	max	min	max	min	max	min	max	min	max
Ortiz-Colberg (1981)	fixed-fixed	C	Stub	8	46.3	71.2	20.8	31.7	6.7	10.3	2.2	2.3	3.4	3.4	0.14	0.50	6.9	24.0
Ortiz-Colberg (1981)	weak axis pinned	C	Long	15	46.2	71.6	20.4	31.7	6.6	10.3	2.3	2.3	7.7	17.9	0.14	0.43	18.0	126
Sivakumaran (1987)	fixed-fixed	C	Stub	12	57.6	118	25.8	32.0	7.9	9.8	2.2	3.7	1.7	2.4	0.18	0.57	2.2	12.1
Miller and Peköz (1994)	fixed-fixed	S,O,R,C	Stub	14	47.9	173	19.3	39.8	6.3	9.1	2.5	4.5	3.0	3.0	0.26	0.45	3.9	8.0
Abdel-Rahman (1997)	fixed-fixed	S,O,R,C	Stub	8	79.9	108	22.1	32.8	6.9	10.3	2.4	4.9	2.1	3.0	0.31	0.38	3.0	6.7
Pu et al. (1999)	fixed-fixed	S	Stub	9	50.0	122	26.0	65.0	8.0	20.0	1.9	1.9	3.7	3.7	0.27	0.27	13.6	13.9
Moen and Schafer (2008)	fixed-fixed	SL	Short	6	91.7	146	37.0	42.8	9.7	11.1	2.2	3.8	4.0	6.7	0.25	0.42	6.0	6.0
Moen and Schafer (2008)	fixed-fixed	SL	Intermediate	6	91.0	140	37.2	41.0	9.7	10.6	2.3	3.8	7.9	13.3	0.25	0.41	12.0	12.1

S=square, O=oval, R=rectangular, C=circular, SL=slotted

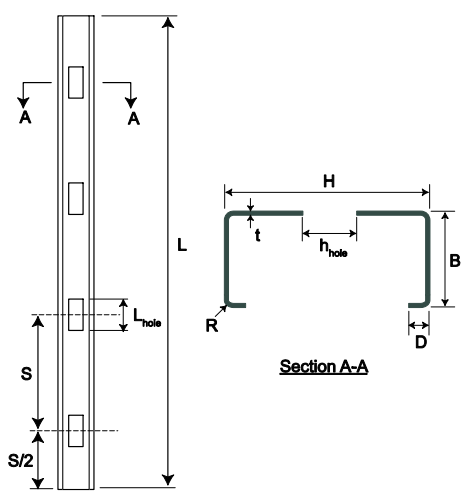

Fig. 3. Column geometry notation

NONLINEAR FE SIMULATIONS OF CFS COLUMNS WITH HOLES

Nonlinear finite element modeling studies of 213 cold-formed steel columns with holes were conducted in ABAQUS to supplement the experimental database in Table 1 and guide the development of the DSM design expressions. Distortional buckling and local-global buckling strength limit states were isolated and explored by combining specific column lengths and column cross-sections from a library of 99 C-section structural studs listed in the Steel Stud Manufacturers Association catalog (SSMA 2001). The modeling protocol utilized herein was developed with care by the authors between 2005 and 2008 (Moen 2008) and validated with experiments on cold-formed steel columns with holes (Moen and Schafer 2008). Nonlinear finite element modeling is a powerful tool for studying the load-deformation response of cold-formed steel members, however results can vary widely with finite element type, mesh density, and solution algorithm (Schafer et al. 2010), assumed initial geometric imperfections (Moen 2008; Schafer and Peköz 1998), the choice of isotropic or kinematic hardening rules (Gao and Moen 2010), and the treatment of through-thickness residual stresses and plastic strains from the cold-work of forming effect (Moen et al. 2008; Quach et al. 2006).

Finite Element Modeling Protocol

Boundary conditions for the 213 simulated column tests were pinned-pinned free-to-warp [Fig. 4(a)]. The columns were meshed with ABAQUS S9R5 thin shell elements employing custom Matlab code which followed the element meshing guidelines described in Moen (2008), where the finite element aspect ratio was specified between 1:1 and 8:1 and at least two elements were provided per local buckling half-wave. A

uniform compressive stress was applied at each end with consistent nodal loads compatible with S9R5 element shape functions (Schafer 1997). The nodal loads were distributed over the first two layers of cross-section nodes to avoid localized failures at the loaded edges. The dimension notation for all SSMA cross-section types is provided in the Appendix. The Appendix and Fig. 3 define the cross-section dimensions, as in, the SSMA 600S162-33 cross-section has H=152.4 mm (6 in.), B=41.1 mm (1.625 in.), D=12.2 mm (0.5 in.), t=0.88 mm (0.0346 in.), and R=2.82 mm (0.111 in.).

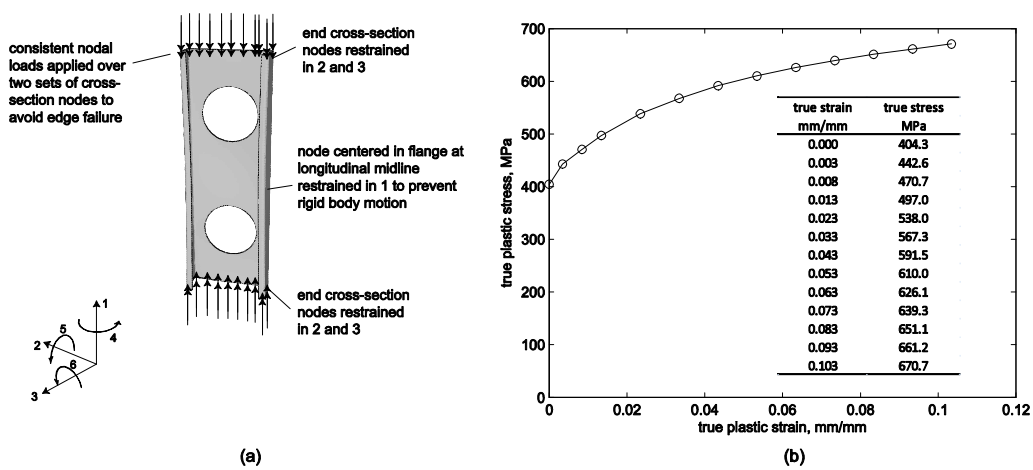

Fig. 4. Nonlinear finite element model (a) boundary conditions and loading, (b) material true plastic stress-strain curve input into ABAQUS

ABAQUS simulations were performed with the modified Riks nonlinear solution algorithm (Crisfield 1981; Powell and Simons 1981; Ramm 1981). Automatic time stepping was enabled with a suggested initial arc length step of 0.25 (note that the Riks method increments in units of energy), a maximum step size of 0.75, and the maximum number of solution increments set at 300. Steel yielding and plasticity were simulated in ABAQUS with isotropic hardening. The same true stress-strain curve was assumed for all column models [Fig. 4(b)], where the steel yield stress F_y=404 MPa, the modulus of elasticity E=203.4 GPa, and Poisson's ratio ν=0.3. Plasticity was initiated in ABAQUS

at the 0.2% yield offset because of recent observations that ABAQUS incorrectly underpredicts column stiffness and ultimate strength when material nonlinearity is initiated at the proportional limit (Moen 2008; Schafer et al. 2010). Residual stresses and initial plastic strains from coiling and corner cold-bending (Moen et al. 2008) were observed to have a minimal influence on load-deformation response when implemented with isotropic hardening (Moen 2008) and were not considered. However, recent evidence suggests that the full load-deformation response of a cold-formed steel column is simulated more accurately with a combined isotropic-kinematic hardening rule and user input through-thickness residual stresses and plastic strains (Gao and Moen 2010).

Initial geometric imperfections were imposed on the column geometry in ABAQUS with custom Matlab code which combines the local, distortional buckling, and global elastic buckling mode shapes from a finite strip analysis (i.e. CUFSM) along the column length. The local and distortional imperfection magnitudes were determined based on the statistical approach developed by Schafer and Peköz (1998) where the probability, P, that a random imperfection magnitude, Δ, is less than a deterministic imperfection, d, is defined with a cumulative distribution function (CDF) derived from measured data. Four simulations were performed for each column, $P(\Delta<d)=0.25$ local and distortional imperfection magnitudes with $L/2000$ global imperfections (where L is the length of the column), and $P(\Delta<d)=0.75$ local and distortional imperfection magnitudes with a global imperfection magnitude of $L/1000$. The global imperfection magnitude assumptions are based on hot-rolled column out-of-straightness measurements (Galambos 1998) because formal guidelines are currently unavailable for cold-formed steel columns, though work is underway (Zeinoddini and Schafer 2008).

The column global imperfection shape was defined by the lowest global buckling mode, either weak-axis flexural buckling or flexural-torsional buckling, depending on the cross-section dimensions and column length. The use of ± L/1000 and ± L/2000 imperfection magnitudes were required when the imperfection shape was weak-axis flexural buckling, because a C-section is singly-symmetric and column capacity varies depending on the imperfection direction, i.e. if flexure places the C-section web in compression or the flange lips in compression. Global imperfections were not considered for columns with $L/H \leq 18$ (i.e., stockier columns with a low sensitivity to global imperfections).

The critical elastic buckling loads, including the influence of holes, for each column considered in the study were calculated based on the simplified methods described in Moen and Schafer (2009a). The local ($P_{cr\ell}$) and distortional (P_{crd}) buckling loads were obtained with finite strip approximate methods, and the global buckling load (P_{cre}) was calculated with the weighted average hand approximation. The complete database of simulated column experiments, including cross-section type, column and hole geometry, simulated ultimate strengths ($\pm P_{sim25}$ and $\pm P_{sim75}$) and critical elastic buckling loads for each column is provided in Moen (2008).

Distortional Buckling Column Failures

A group of 20 SSMA columns from the column simulation database (Moen 2008, Appendix K, Study Type D) were chosen to evaluate the influence of P_{ynet}/P_y on the simulated strength of columns, P_{sim25} and P_{sim75}, predicted to collapse with a distortional buckling failure mode. (The SSMA cross-section notation is described in the Appendix.) Remember that P_{sim25} and P_{sim75} are simulated strengths of columns with imposed local

and distortional buckling imperfection shapes with two different magnitudes corresponding to $P(\Delta<d)$=0.25 and $P(\Delta<d)$=0.75 respectively. Global imperfections are not considered for these relatively stocky columns. The column length is L=610 mm (24 in.) which ensures that at least two distortional buckling half-waves form in a column. The cross-sections selected for this study (Table 2) were chosen to have relatively thick sheet steel (t up to 2.58 mm) to avoid a local buckling-controlled failure. The web of each column has two circular holes where the hole spacing S=305 mm (12 in.). The hole depth (diameter), h_{hole}, is varied for each column to produce P_{ynet}/P_y of 1.0 (no holes), 0.80, and 0.60.

Table 2. Distortional buckling failure mode study

		Elastic buckling (including holes)			Column capacity					
		P_{crd} (kN)			P_{sim25} (kN)			P_{sim75} (kN)		
SSMA		P_{ynet}/P_y			P_{ynet}/P_y			P_{ynet}/P_y		
Cross-Section	P_y (kN)	1.00	0.80	0.60	1.00	0.80	0.60	1.00	0.80	0.60
600S250-97	304	280	262	242	224	195	135	208	184	137
600S162-97	251	209	190	171	172	153	114	151	135	113
800S250-97	357	202	186	170	217	205	163	204	176	165
800S200-97	331	171	156	141	181	185	141	165	178	133
800S162-97	304	131	117	102	156	152	126	143	141	120
600S137-68	167	66.2	59.1	51.8	81.4	80.5	66.3	73.8	72.9	63.6
1000S250-97	411	139	126	113	198	200	156	192	182	153
1000S200-97	384	114	102	89.7	197	173	150	189	162	150
800S162-68	218	56.7	51.7	46.5	102	90.3	83.2	111	85.4	85.8
1000S162-97	357	88.6	76.9	64.9	157	157	134	150	178	132
600S137-43	108	24.4	22.6	20.6	42.5	42.4	34.2	41.2	44.3	34.8
1200S250-97	464	97.9	87.4	76.5	203	177	163	179	161	150
1000S200-68	273	50.2	45.9	41.5	97.4	102	86.3	91.2	99.2	86.7
1000S250-54	233	38.9	36.4	33.8	73.8	71.2	68.1	69.8	68.1	67.6
1200S162-97	411	65.6	55.0	43.9	154	153	139	157	157	144
1000S162-68	255	36.9	33.0	28.9	89.0	87.6	79.6	88.1	83.2	78.7
800S162-43	140	19.7	18.3	16.9	45.8	44.5	39.4	44.5	44.5	39.1
1200S250-68	329	42.9	NR	35.3	97.4	NR	97.4	110	NR	88.5
1200S200-68	311	37.3	34.0	30.6	108	98.3	93.4	101	101	81.8
1000S200-43	175	19.5	18.3	17.1	49.4	46.7	45.4	48.5	48.9	45.4

NR=FE model did not converge

P_{crd} was calculated including the influence of holes with the FSM approach described in Moen and Schafer (2009a)

The column strengths, P_{sim25} and P_{sim75}, diverge from the DSM prediction curve as distortional slenderness, λ_d, decreases as shown in Fig. 5(a) and Fig. 5(b). When λ_d is high (i.e. P_{crd} is low relative to P_y), the column strength is controlled by elastic buckling, and the influence of holes on strength is reflected in the reduction in P_{crd} and the resulting increase in λ_d. When λ_d is low, column failure is initiated by inelastic buckling and yielding of the cross-section at the location of a hole (i.e., at the net section) resulting in the collapse of the unstiffened strips adjacent to the hole. The transition from an elastic buckling-dominated failure to a failure initiated by yielding and collapse of the net section is presented for an SSMA 800S250-97 structural stud in Fig. 6.

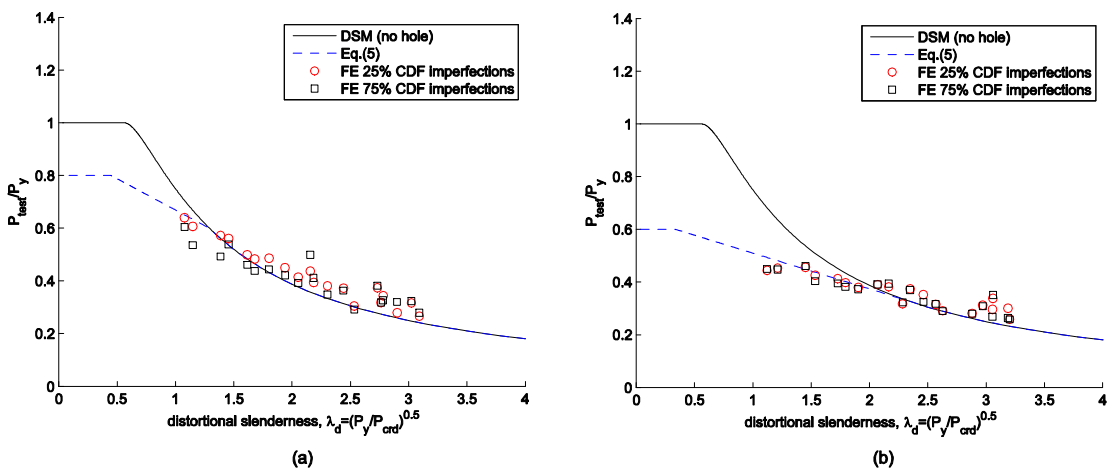

Fig. 5. Column strengths fall below DSM predictions as hole size and slenderness decreases for distortional buckling controlled failures: (a) P_{ynet}/P_y=0.80, (b) P_{ynet}/P_y=0.60

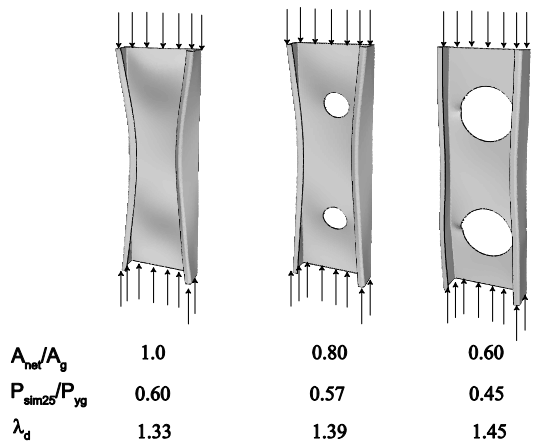

Fig. 6. SSMA 800S250-97 structural stud failure mode transition from distortional buckling to yielding at the net section

Global Buckling Column Failures

A group of 18 columns predisposed to global buckling failure were selected from the simulation database (Moen 2008, Appendix K, Study Type G) for this study. The column length, L, varied from 200 mm to 1152 mm resulting in columns with a range of global column slenderness, $\lambda_c=(P_y/P_{cre})^{0.5}$ between 0.30 and 3.6. The cross-sections in this study (see Table 3) were chosen to avoid a local buckling or distortional buckling-controlled failures. The web of each column contains evenly spaced slotted holes where the hole spacing S varies from 203 mm (8 in.) to 559 mm (22 in.). The hole length, L_{hole}, is held constant at 102 mm (4 in.), while the hole depth, h_{hole}, is varied for each column to produce P_{ynet}/P_y of 1.0 (no holes), 0.90, and 0.80. The first four columns in Table 3 were modeled with circular holes instead of slotted holes because the slotted holes resulted in impractical column layouts, with the hole extending over more than 50% of the column length. The global imperfection shape for five of the longer columns was weak-axis flexural buckling, and therefore four simulated strengths, $P_{sim25^{\pm}}$ and $P_{sim75^{\pm}}$, are calculated for these columns.

Table 3. Global buckling failure mode study

SSMA Cross-Section	L (mm)	S (mm)	P_y (kN)	Elastic buckling (including holes) P_{cre} (kN) P_{ynet}/P_y 1.00	0.90	0.80	Column capacity P_{sim25} (kN) P_{ynet}/P_y 1.00	0.90	0.80	P_{sim75} (kN) P_{ynet}/P_y 1.00	0.90	0.80
250S162-68*	203	203	115	1349	1292	NR	110	107	NR	95.6	94.7	NR
250S137-68*	305	305	102	443	425	395	91.6	88.5	75.2	76.1	75.6	70.3
250S162-68*	406	406	115	344	331	308	102	105	85.8	95.2	94.3	85.8
250S162-68*	559	559	115	186	179	168	101	102	81.0	88.5	88.1	79.2
250S137-54	660	330	82.2	82.5	79.2	74.0	59.6	54.7	47.1	47.6	45.4	42.3
250S137-54	813	406	82.2	56.4	54.3	50.9	46.7	45.4	43.2	39.8	38.5	37.9
400S162-68	1372	330	143	72.9	70.2	65.7	64.9	64.1	58.7	56.9	53.4	50.7
600S250-97	2337	330	304	119	113.9	105.4	104	103	103	92.5	91.2	91.2
350S162-54	1676	330	108	33.3	NR	30.1	31.1	NR	30.2	29.0	NR	27.9
250S162-33	1473	356	58.2	14.5	14.0	13.1	13.3	12.9	12.7	12.1	11.7	11.5
250S137-33	1524	305	51.4	10.6	10.22	9.56	10.1	9.87	9.61	9.56	9.21	9.07
362S137-43	2134	305	79.6	13.8	13.28	12.49	12.3/11.6	12.3/12.3	11.7/11.7	10.7/11.0	10.5/10.7	11.6/10.5
362S137-68	2235	305	122	18.1	18.1	17.8	17.3/17.7	17.1/17.0	16.7/16.1	14.5/15.3	14.2/14.8	17.7/14.4
250S162-54	2438	305	93.3	11.9	11.60	11.00	11.9	11.6	11.3	11.6	11.2	10.9
600S137-54	2438	305	134	14.7	14.7	14.5	14.5/13.1	14.3/12.6	13.5/12.3	13.7/11.7	12.9/11.3	13.1/11.3
250S137-33	2388	330	51.4	5.08	4.92	4.64	4.98	4.89	4.76	4.89	4.71	4.6
800S137-97	2388	330	285	24.8	24.7	24.5	23.7/22.3	23.0/21.8	22.6/21.5	21.7/20.0	21.5/19.7	22.3/19.6
800S137-97	2438	305	285	23.8	23.7	23.5	22.6/21.4	22.0/20.9	21.6/20.5	20.6/19.1	20.5/18.9	21.4/18.7

* column with a single circular hole

XX/XX=P_{sim25+}/P_{sim25-} or P_{sim75+}/P_{sim75-}

NR=FE model did not converge

P_{cre} was calculated including the influence of holes with the simplified approach described in Moen and Schafer (2009a)

Fig. 7(a) and Fig. 7(b) demonstrate that the simulation results trend with the predictions in the elastic buckling regime (λ_c >1.5), but diverge below the DSM prediction curve as λ_c decreases and P_{ynet}/P_y increases. The columns with strengths falling below DSM predictions range in length from 20 mm to 66 mm with a low global slenderness. In these cases, strength is limited by the capacity of the net section, which is consistent with the distortional buckling failure study (Fig. 6).

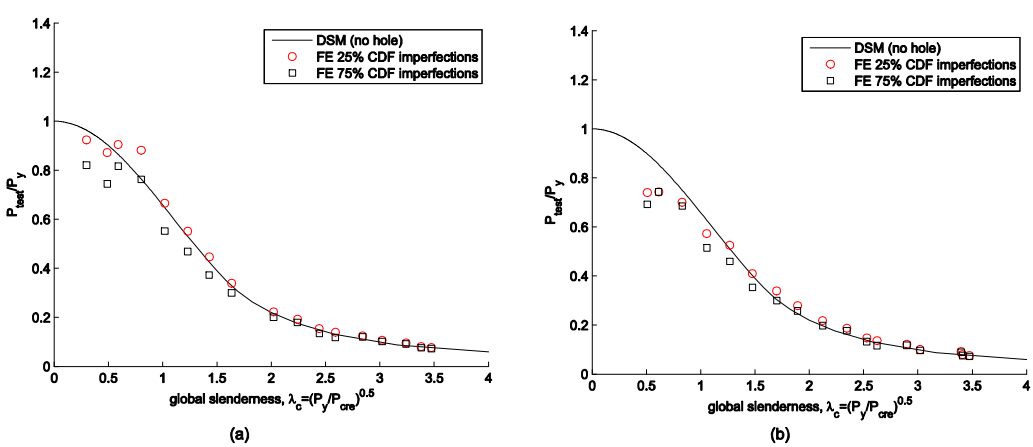

Fig. 7 Column strengths fall below DSM predictions as hole size and slenderness decreases for global buckling controlled failures: (a) P_{ynet}/P_y=0.90, (b) P_{ynet}/P_y=0.80

Local-Global Buckling Interaction Column Failures

The distortional buckling and global buckling studies demonstrate when slenderness is high, i.e. when elastic buckling dominates column failure, that the critical elastic buckling loads, calculated including the influence of holes, can be used with the existing DSM design expressions to accurately predict ultimate strength. When slenderness is low, inelastic buckling and yielding at a hole limit column strength to P_{ynet}. The goal of this study is to determine if the same trends apply for columns with holes failing by local-global buckling interaction.

A group of 11 columns predisposed to local-global buckling interaction were selected from the simulation database (Moen 2008, Appendix K, Study Type L) for this study (Table 4). The columns have SSMA cross-sections and lengths which result in a local buckling slenderness, λ_ℓ, ranging from 0.8 to 3.0. The column length, L, varies from 610 mm (24 in.) to 2235 mm (88 in.) and column widths range from 89 mm (3.5 in.) to 305 mm (12 in.). The web of each column contains evenly spaced circular holes where the hole spacing S varies from 305 mm (12 in.) to 432 mm (17 in.). The hole depth (diameter), h_{hole}, is varied for each column to produce P_{ynet}/P_y of 1.0 (no holes), 0.80, and 0.65.

Table 4. Local-global buckling interaction failure mode study

				Elastic buckling (including holes)						Column capacity					
				$P_{cr\ell}$ (kN)			P_{cre} (kN)			P_{sim25} (kN)			P_{sim75} (kN)		
				P_{ynet}/P_y			P_{ynet}/P_y			P_{ynet}/P_y			P_{ynet}/P_y		
SSMA Cross-Section	L (mm)	S (mm)	P_y (kN)	1.00	0.80	0.65	1.00	0.80	0.65	1.00	0.80	0.65	1.00	0.80	0.65
350S162-68	864	204	134	149	149	149	140	126	104	22.3/19.3	19.3/18.3	13.0/13.0	18.7/18.7	17.4/17.4	12.7/12.7
1000S200-97	2235	144	384	105	105	105	102	100	94.2	19.4/17.0	17.9/16.3	16.2/15.2	18.0/15.5	17.3/15.2	16.2/14.5
350S162-54	610	144	108	74.4	74.4	74.4	220	195	159	16.9/16.9	15.1/15.1	11.2/11.2	15.0/15.0	14.3/14.3	11.0/11.0
800S200-68	1880	144	236	50.7	50.7	50.7	103	100	88.0	14.0/14.7	13.5/14.3	12.2/12.9	13.2/14.5	13.0/14.3	11.8/13.1
550S162-54	1067	168	138	39.1	39.1	39.1	129	119	98.8	13.2/12.7	13.0/12.5	10.9/10.4	12.5/11.6	12.3/11.5	10.6/9.86
800S200-54	1676	156	189	25.6	25.6	25.6	106	104	88.3	11.3/11.8	11.0/11.5	9.97/10.4	11.0/11.9	10.7/11.6	9.85/10.6
600S250-43	1422	168	140	20.4	20.4	20.4	132	116	92.8	12.0/12.3	11.6/11.9	9.51/9.77	11.8/12.2	11.4/11.9	9.01/9.93
600S162-43	813	192	116	17.7	17.7	17.7	187	180	151	10.2/10.0	10.2/9.98	8.83/8.77	10.3/9.97	10.1/9.89	9.04/8.72
800S250-43	1880	144	163	13.7	13.7	13.7	111	97.1	77.2	9.84/9.62	9.38/NR	8.87/8.58	9.90/9.40	9.71/9.19	8.90/8.39
800S162-43	1016	156	140	12.1	12.1	12.1	129	127	122	NR/9.32	8.46/9.14	7.89/NR	8.74/9.42	8.68/9.25	8.18/6.88
1000S250-43	2032	156	187	10.2	10.2	10.2	107	105	84.5	8.82/8.79	8.14/8.69	NR/8.11	9.64/8.72	8.84/8.62	8.00/7.46

XX/XX=P_{sim25+}/P_{sim25-} or P_{sim75+}/P_{sim75-}

NR = FE model did not converge

P_{cre} and $P_{cr\ell}$ were calculated including the influence of holes with the simplified methods described in Moen and Schafer (2009a)

Fig. 8 demonstrates that the simulated strengths, $P_{sim25\pm}$ and $P_{sim75\pm}$, for the 11 columns are in most cases consistent with the DSM predicted strength, $P_{n\ell}$. The decreasing trend in column strength with increasing hole size is accurately predicted because P_{cre} is calculated including the influence of holes, causing λ_c to increase as P_{ynet}/P_y decreases. The local buckling load, $P_{cr\ell}$, is unaffected by the presence of circular holes (see Table 4) because the unstiffened strips adjacent to the hole are predicted to buckle at a higher axial force than the gross cross section between holes (Moen and Schafer 2009a).

Two isolated cases, the SSMA 350S162-68 and 1000S200-97 columns, exhibit disproportionate strength reductions as the hole size increases to P_{ynet}/P_y =0.65 (Fig. 8) caused by changes in the global buckling failure mode from the presence of holes. For example, when P_{ynet}/P_y=0.80, the SSMA 350S162-68 column fails in flexural-torsional buckling with a 12% strength reduction, while for the same column with larger holes (P_{ynet}/P_y=0.65), collapse of the net section results in a weak-axis flexural failure and a 42% strength reduction. A change in global buckling mode caused by the presence of holes has not been documented in existing experimental literature, and is hypothesized to result from the idealized warping free end conditions assumed in the simulations. Future experimental work is needed on cold-formed steel columns with intermediate global slenderness and large holes to determine if global mode switching caused by the presence of holes should be considered in design.

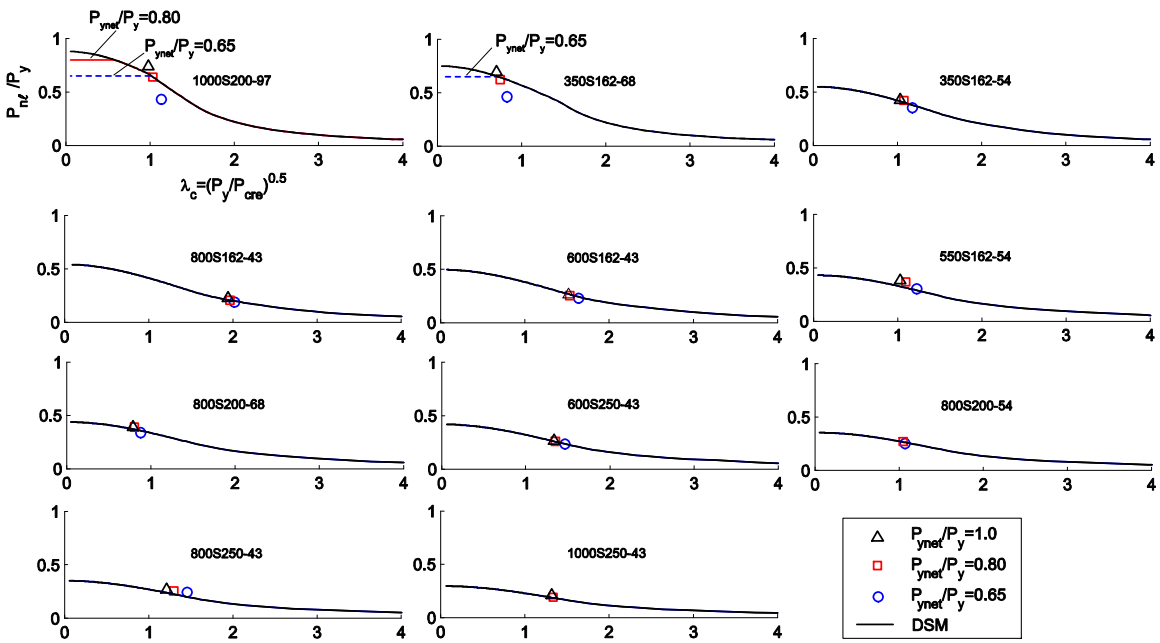

Fig. 8. Column strengths governed by local-global buckling interaction trend with DSM design curves (simulation results shown are $P_{test25+}$, $P_{test25-}$ and $P_{test75\pm}$ are similar)

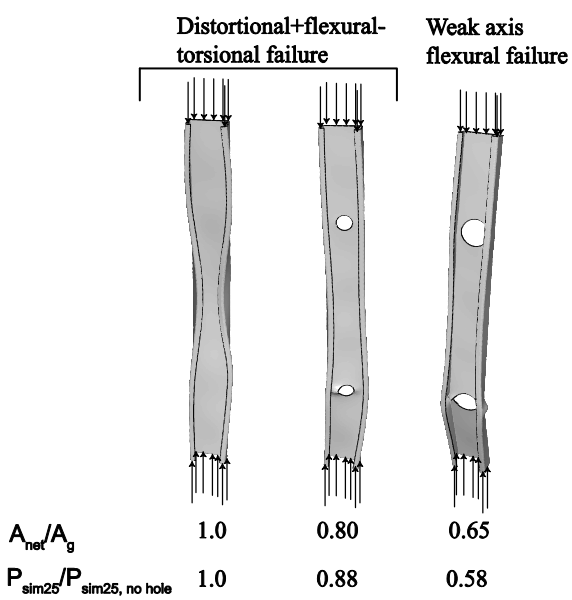

Fig. 9. SSMA 350S162-68 column failure mode switches from a flexural-torsional buckling failure to weak axis flexure as hole size increases

DESIGN METHOD DEVELOPMENT

The nonlinear finite element studies presented in the previous section confirm that the existing DSM design expressions are viable for cold-formed steel columns with holes when the failure mode is controlled by elastic buckling, but that modifications are needed in the inelastic regime. Several options for modifying the existing DSM expressions are presented and evaluated in the following section, with the AISI Main Specification (effective width) serving as a baseline for comparison.

AISI Main Specification

The AISI-S100-07 Main Specification considers two limit states for cold-formed steel columns, (1) local-global buckling interaction (AISI-S100-07, Section C4.1) and (2) distortional buckling (AISI-S100-07, Section C4.2). Column strength is taken as the minimum of P_n (local-global buckling interaction) and P_{nd} (distortional buckling). Column strength predictions for the local-global buckling limit state are calculated with the equation:

$$P_n = A_e F_n \,, \tag{4}$$

where A_e is the column's effective area and F_n is the global column strength (stress). The effective width of a stiffened element (e.g., a C-section web) containing non-circular web holes is calculated with the unstiffened strip approach (Miller and Peköz 1994). The reduction in effective width from circular holes is obtained with empirical equations derived from experiments (Ortiz-Colberg 1981) . The column strength, F_n, is the stress equivalent to P_{ne} in DSM, i.e. $P_{ne}=F_n A_g$, except that critical elastic global buckling stress

F_e=P_{cre}/A_g in the global slenderness term, λ_c=$(F_y/F_e)^{0.5}$, does not include the influence of holes. (This is a fundamental difference between the proposed DSM approach for columns with holes and the AISI Main Specification.) The effective area is limited to the net cross-sectional area, A_{net}, which restricts the column strength to P_{ynet}. The distortional buckling column strength, $P_{nd,}$ is predicted in the Main Specification with DSM expressions, where the distortional buckling load, P_{crd}, is calculated ignoring holes (again fundamentally different than the proposed DSM approach in this manuscript).

DSM Option 1 – use existing DSM equations

The critical elastic buckling loads $P_{cr\ell}$, P_{crd}, and P_{cre} are calculated including the influence of holes, otherwise the existing DSM expressions in Eqs. (1)-(3) are unchanged.

DSM Option 2 – replace P_y with P_{ynet} in all DSM equations

The column squash load, P_y, is replaced with P_{ynet} in Eqs. (1)-(3). The critical elastic buckling loads $P_{cr\ell}$, P_{crd}, and P_{cre} are calculated including the influence of holes.

DSM Option 3 – limit $P_{n\ell}$ and P_{nd} to P_{ynet}

The distortional buckling capacity, P_{nd}, and the local buckling (local-global buckling interaction) capacity, $P_{n\ell}$, in Eqs. (2)-(3) are limited to P_{ynet}. The critical elastic buckling loads $P_{cr\ell}$, P_{crd}, and P_{cre} are calculated including the influence of holes.

DSM Option 4 – limit $P_{n\ell}$ to P_{ynet} and transition P_{nd} to P_{ynet}

The local buckling (local-global buckling interaction) capacity, $P_{n\ell}$, in Eq. (3) is limited to P_{ynet}, and a modified DSM distortional buckling strength curve (see Fig. 5)

replaces Eq. (2) with a transition from elastic buckling to yielding at the net cross-section:

$$\text{for } \lambda_d \le \lambda_{d1},\ P_{nd} = P_{ynet}$$

$$\text{for } \lambda_{d1} < \lambda_d \le \lambda_{d2},\ P_{nd} = P_{ynet} - \left(\frac{P_{ynet} - P_{d2}}{\lambda_{d2} - \lambda_{d1}}\right)(\lambda_d - \lambda_{d1})$$

$$\text{for } \lambda_d > \lambda_{d2},\ P_{nd} = \left(1 - 0.25\left(\frac{P_{crd}}{P_y}\right)^{0.6}\right)\left(\frac{P_{crd}}{P_y}\right)^{0.6} P_y, \qquad (5)$$

where λ_{d1}=0.561(P_{ynet}/P_y), λ_{d2}=0.561[14$(P_y/P_{ynet})^{0.4}$-13], and

$$P_{d2} = \left(1 - 0.25\left(\frac{1}{\lambda_{d2}}\right)^{1.2}\right)\left(\frac{1}{\lambda_{d2}}\right)^{1.2} P_y. \qquad (6)$$

The critical elastic buckling loads $P_{cr\ell}$, P_{crd}, and P_{cre} are calculated including the influence of holes.

DSM Option 5 – Limit P_{ne} to P_{ynet}, transition P_{nd} to P_{ynet}

The global buckling capacity, P_{ne}, in Eq. (1) is limited to P_{ynet}, and the modified DSM distortional buckling strength curve in Eq. (5) replaces Eq. (2). The critical elastic buckling loads $P_{cr\ell}$, P_{crd}, and P_{cre} are calculated including the influence of holes.

DSM Option 6 – Transition P_{ne} to P_{ynet}, transition P_{nd} to P_{ynet}

The global buckling capacity, P_{ne}, in Eq. (1) is replaced with the following expression that provides a transition from the elastic portion of the global buckling strength curve to P_{ynet}:

$$\text{for } \lambda_c \le 1.5,\ P_{ne} = \left(0.658^{\lambda_c^2}\right) P_{ynet}\left(\frac{P_y}{P_{ynet}}\right)^{(\lambda_c/1.5)^2} \le P_{ynet}$$

$$\text{for } \lambda_c > 1.5,\ P_{ne} = \left(\frac{0.877}{\lambda_c^2}\right) P_y \le P_{ynet} \tag{7}$$

The modified DSM distortional buckling strength curve in Eq. (5) replaces Eq. (2), and the critical elastic buckling loads $P_{cr\ell}$, P_{crd}, and P_{cre} are calculated including the influence of holes.

PERFORMANCE OF DSM DESIGN EXPRESSIONS

The experimental and simulation databases presented earlier in the manuscript contain useful column test data and elastic buckling properties that are employed in this section to evaluate the 6 DSM options for columns with holes. Test-to-predicted statistics are calculated for each option, and a first order second moment reliability analysis is performed to obtain the LRFD strength reduction factor, ϕ, with the following equation from Chapter F of the AISI Main Specification (AISI-S100 2007):

$$\phi = C_\phi \left(M_m F_m P_m\right) e^{-\beta_o \sqrt{V_M^2 + V_F^2 + C_P V_P^2 + V_Q^2}} . \tag{8}$$

The LRFD calibration coefficient is C_ϕ=1.52, the mean value of the material factor is M_m=1.10 for concentrically loaded compression members, and the mean value of the fabrication factor is F_m=1.0. The professional factor P_m is taken as the test-to-predicted mean, the coefficient of variation (COV) of the material factor is V_m=0.10, the COV of the fabrication factor is V_f=0.05, the COV of the load effect is V_q=0.21 for LRFD, and the correction factor C_p=1. The COV of the test results, V_p, is calculated as the ratio of the standard deviation to the mean of the test-to-predicted statistics.

The experimental test-to-predicted statistics are summarized in Table 5 for all columns and Table 6 for stub columns ($\lambda_c \le 0.20$). The best performing DSM option is

identified as Option 4 – cap $P_{n\ell}$ and transition P_{nd}, with its test-to-predicted nearest unity and low COV of 0.07 and 0.09 for local-global buckling interaction and distortional buckling failures, respectively. The need for a limit on column strength to P_{ynet} is reiterated with the stub column results for DSM Option 1 (Table 6), where the test-to-predicted mean is below unity for all limit states. DSM Option 2 predictions are conservative because the method reduces $P_{n\ell}$ and P_{nd} over all slenderness values instead of just low slenderness values where inelastic buckling and yielding at the net cross section control. DSM Options 5 and 6 are conservative because they unduly penalize the strength of columns with holes controlled by elastic local-global buckling interaction. DSM Options 3 and 4 impose a strength penalty only when the local and global slenderness are both low enough to elicit a yielding failure at the net cross-section. The conclusion that DSM Option 4 is the best performing DSM option is supported by the simulation database test-to-predicted statistics in Table 7.

The AISI effective width method strength predictions across the experiment database (Table 5) are conservative when compared to DSM Option 4 (1.17 vs. 1.07) with a higher COV (0.09 vs. 0.07) than DSM Option 4 for local-global buckling interaction failures. The AISI Main Specification is applicable to only 23 of the 59 test specimens controlled by local-global buckling interaction because of limits on hole geometry, while DSM is applicable to all specimens considered. The simulation database statistics demonstrates that DSM is accurate over a wide range of hole sizes, hole spacings, and cross-section dimensions, while the Main Specification is unconservative across the full data set as evidenced by the 0.91 test-to-predicted mean for distortional buckling controlled failures.

Table 5. Experimental database test-to-predicted statistics

Method	Option	Description	Controlling Limit State											
			Local-global interaction				Distortional buckling				Global buckling or yielding			
			Mean	COV	n	ϕ	Mean	COV	n	ϕ	Mean	COV	n	ϕ
DSM	1	existing DSM equations (P_y everywhere)	1.03	0.11	52	0.90	1.09	0.15	15	0.90	1.06	0.16	11	0.86
	2	P_{ynet} everywhere	1.17	0.08	47	1.05	1.22	0.11	15	1.06	1.19	0.13	12	1.00
	3	Cap $P_{n\ell}$, cap P_{nd}	1.07	0.07	42	0.96	1.15	0.09	12	1.02	1.03	0.17	13	0.83
	4	Cap $P_{n\ell}$, transition P_{nd}	1.07	0.07	40	0.96	1.11	0.09	31	0.98	1.08	0.19	7	0.84
	5	Cap P_{ne}, transition P_{nd}	1.15	0.07	56	1.03	1.17	0.09	15	1.04	1.11	0.14	7	0.93
	6	Transition P_{ne}, transition P_{nd}	1.17	0.07	57	1.05	1.17	0.09	11	1.04	1.18	0.14	10	0.99
AISI effective width method		all data	1.11	0.09	59	0.98	1.11	0.05	15	1.01	1.35	0.08	4	1.21
		within spec limits	1.17	0.10	23	1.03	**	**	**	**	1.35	0.08	4	1.21
		outside spec limits	1.06	0.06	36	0.96	**	**	**	**	---	---	0	---

** Code limits on hole geometry are not provided for the distortional buckling limit state in AISI-S100-07

Table 6. Experimental database test-to-predicted statistics, stub columns only ($\lambda_c \leq 0.20$)

Method	Option	Description	Controlling Limit State											
			Local-global interaction				Distortional buckling				Global buckling or yielding			
			Mean	COV	n	ϕ	Mean	COV	n	ϕ	Mean	COV	n	ϕ
DSM	1	existing DSM equations (P_y everywhere)	0.98	0.10	33	0.86	0.83	0.01	3	0.76	0.84	0.10	3	0.74
	2	P_{ynet} everywhere	1.12	0.07	28	1.02	1.03	0.05	3	0.94	1.07	0.12	8	0.92
	3	Cap $P_{n\ell}$, cap P_{nd}	1.03	0.06	23	0.94	---	---	0	---	0.79	0.15	16	0.66
	4	Cap $P_{n\ell}$, transition P_{nd}	1.04	0.06	21	0.94	1.08	0.10	16	0.96	0.80	---	2	---
	5	Cap P_{ne}, transition P_{nd}	1.12	0.07	28	1.02	1.14	0.08	9	1.03	0.90	---	2	---
	6	Transition P_{ne}, transition P_{nd}	1.12	0.07	28	1.02	1.14	0.08	9	1.03	0.91	---	2	---
AISI effective width method		all data	1.07	0.06	38	0.97	1.03	---	1	---	---	---	0	---
		within spec limits	1.12	0.05	9	1.02	**	**	**	**	---	---	0	---
		outside spec limits	1.05	0.06	29	0.95	**	**	**	**	---	---	0	---

** Code limits on hole geometry are not provided for the distortional buckling limit state in AISI-S100-07

Table 7. Simulation database test-to-predicted statistics

Method	Option	Description	Controlling Limit State											
			Local-global interaction				Distortional buckling				Global buckling or yielding			
			Mean	COV	n	ϕ	Mean	COV	n	ϕ	Mean	COV	n	ϕ
DSM	1	existing DSM equations (P_y everywhere)	1.00	0.11	95	0.87	1.07	0.16	180	0.88	0.98	0.11	110	0.85
	2	P_{ynet} everywhere	1.10	0.10	88	0.96	1.23	0.15	183	1.02	1.03	0.08	114	0.92
	3	Cap $P_{n\ell}$, cap P_{nd}	1.00	0.11	95	0.87	1.09	0.14	170	0.91	0.95	0.15	120	0.79
	4	Cap $P_{n\ell}$, transition P_{nd}	1.01	0.09	91	0.89	1.07	0.13	202	0.91	1.01	0.09	92	0.89
	5	Cap P_{ne}, transition P_{nd}	1.03	0.09	105	0.91	1.08	0.14	188	0.91	1.01	0.09	92	0.89
	6	Transition P_{ne}, transition P_{nd}	1.07	0.09	153	0.94	1.13	0.13	120	0.96	1.02	0.08	112	0.91
AISI effective width method		all data	1.03	0.11	150	0.89	0.91*	0.09	149	0.80	1.13	0.11	86	0.98
		within spec limits	1.03	0.09	18	0.91	**	**	**	**	1.09	0.11	6	0.95
		outside spec limits	1.03	0.12	132	0.89	**	**	**	**	1.13	0.11	80	0.98

* low test-to-predicted ratio results from inaccurate L-G and D buckling limit state predictions

** Code limits on hole geometry are not provided for the distortional buckling limit state in AISI-S100-07

CONCLUSIONS

The infrastructure is now in place to extend the AISI Direct Strength Method to cold-formed steel columns with holes. Elastic buckling properties including the influence of holes can be conveniently obtained with general, accessible hand methods and new procedures utilizing the finite strip method. The local, distortional, and global

buckling loads, calculated including holes, work in harmony with the existing DSM design expressions to accurately predict column strength when cross-section or global slenderness is high and elastic buckling controls the failure mode. Modifications were required to the DSM design equations in the inelastic regime, as the net cross section at a hole becomes the weak point and limits column capacity.

The recommended DSM equations for columns with holes were validated with a broad data set of tested column strengths, including existing experiments and nonlinear finite element simulations performed with a validated modeling protocol. The proposed DSM distortional buckling strength prediction equations provide a transition from the elastic buckling failure regime to the net section strength limit. The DSM predicted local-global buckling interaction strength is limited to the capacity at the net section for the case when both global and local slenderness are low, but otherwise remains unchanged. The recommended DSM design equations were demonstrated to be viable across a wide range of hole sizes, shapes, spacings, and column dimensions, outperforming the AISI Main Specification from the perspective of accuracy and generality.

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APPENDIX

The following table relates the SSMA cross-section notation to column dimensions employed in this manuscript (see Fig. 3).

Thickness	t (mm)	R (mm)	Section	B (mm)	D (mm)
33	0.88	2.82	S125	31.8	4.8
43	1.15	2.95	S137	34.9	9.5
54	1.44	3.59	S162	41.3	12.7
68	1.81	4.53	S200	50.8	15.9
97	2.58	6.46	S250	63.5	15.9